STUDENT PROJECT REPORT

PowerScope

ESP32 energy monitor for single-phase 230 V loads up to 6 A

Submitted by

Team members _____

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Department / Class _____

College / Institution _____

Project guide _____

Academic year _____

> **Mains voltage**
> This project connects to 230 V AC. Work on it only when it is unplugged, keep every module inside a closed box, and never connect USB while it is on mains.

1. Introduction

PowerScope is a plug-in energy monitor built into a modular wall-plate box. A load (fan, cooler, kettle, charger) plugs into its front socket. PowerScope measures the mains voltage and the load current many thousand times a second and works out voltage, current, real, reactive and apparent power, power factor, phase angle, crest factor, frequency and energy. It shows them on a small OLED screen and on a dashboard served over its own WiFi hotspot, so any phone can open it without the internet.

Aim: to build a low-cost, safe, self-contained instrument that makes the electrical behaviour of everyday loads visible, for learning and demonstration.

2. Specifications

Table 1. PowerScope specifications

Item	Value
Supply and load	230 V AC, 50 Hz, single phase; load up to 6 A (about 1.3 kW)
Protection	single-pole MCB; 1 A slow-blow fuse on the internal supply
Measured	Vrms, Irms, real power P (W), reactive power Q (var), apparent power S (VA), power factor, phase angle, crest factor, frequency, energy (kWh), load-on time, peak power
Sampling	1600 samples per channel in 200 ms (8 kHz), 10 mains cycles per window; new values 5 times a second
Voltage sensor	ZMPT101B transformer module, through a 10k / 20k divider into GPIO34
Current sensor	ACS712-20A Hall sensor (100 mV/A), in series with the live wire, divider into GPIO35 (66.7 mV/A at the pin)
Accuracy	a few percent after calibration against a multimeter and a clamp meter
Display	1.3" OLED SH1106, 128 x 64, I2C; six pages that change every 5 s
Web	own WiFi hotspot "PowerScope" (password powerscope), http://192.168.4.1; live values, trends, scope, energy and cost, calibration
Data kept after power-off	energy total, 62 days and 12 months of kWh, tariff, calibration, PIN (ESP32 flash, saved at most every 10 minutes)
Power use	about 1 W for the electronics, from a Hi-Link HLK-5M05 (5 V, 1 A)

3. How it works

Figure 1 (next page) shows the blocks. The power path is on top: mains, MCB, the current sensor in the live wire, the rocker switch and the socket. Two taps after the MCB feed the voltage sensor and the 5 V supply. Both sensor outputs go through a divider and filter into the ESP32, which does all the maths and drives the screen and the web page.

3.1 Sensing

- **Voltage:** the ZMPT101B is a small transformer with an op-amp. It gives an AC wave around about 2.5 V, about 1 mV at its output per volt of mains (as trimmed).
- **Current:** the ACS712-20A sits in the live wire. Its output is 2.5 V at zero current and moves 100 mV per amp.
- **Divider:** both modules run on 5 V, but the ESP32 input must stay below 3.3 V. A 10 kΩ resistor from the sensor output to the input pin and 20 kΩ from the pin to ground scale the signal by 2/3. The 100 nF capacitor across the 20 kΩ removes high-frequency noise.
- **ADC:** GPIO34 and GPIO35 are on ADC1, which keeps working while WiFi is on (ADC2 does not).

3.2 Maths, every 200 ms

1. Vrms and Irms are the root of the mean square. Real power P is the mean of v × i. S = Vrms × Irms, PF = |P| / S, Q = $\sqrt{(S^2 - P^2)}$, phase angle = acos(PF), crest factor = peak current / Irms.
2. Frequency comes from the rising zero crossings of the voltage (with hysteresis).
3. Energy adds P × time. Days and months use the phone's clock, sent by the web page when it opens (the ESP32 has no battery clock).

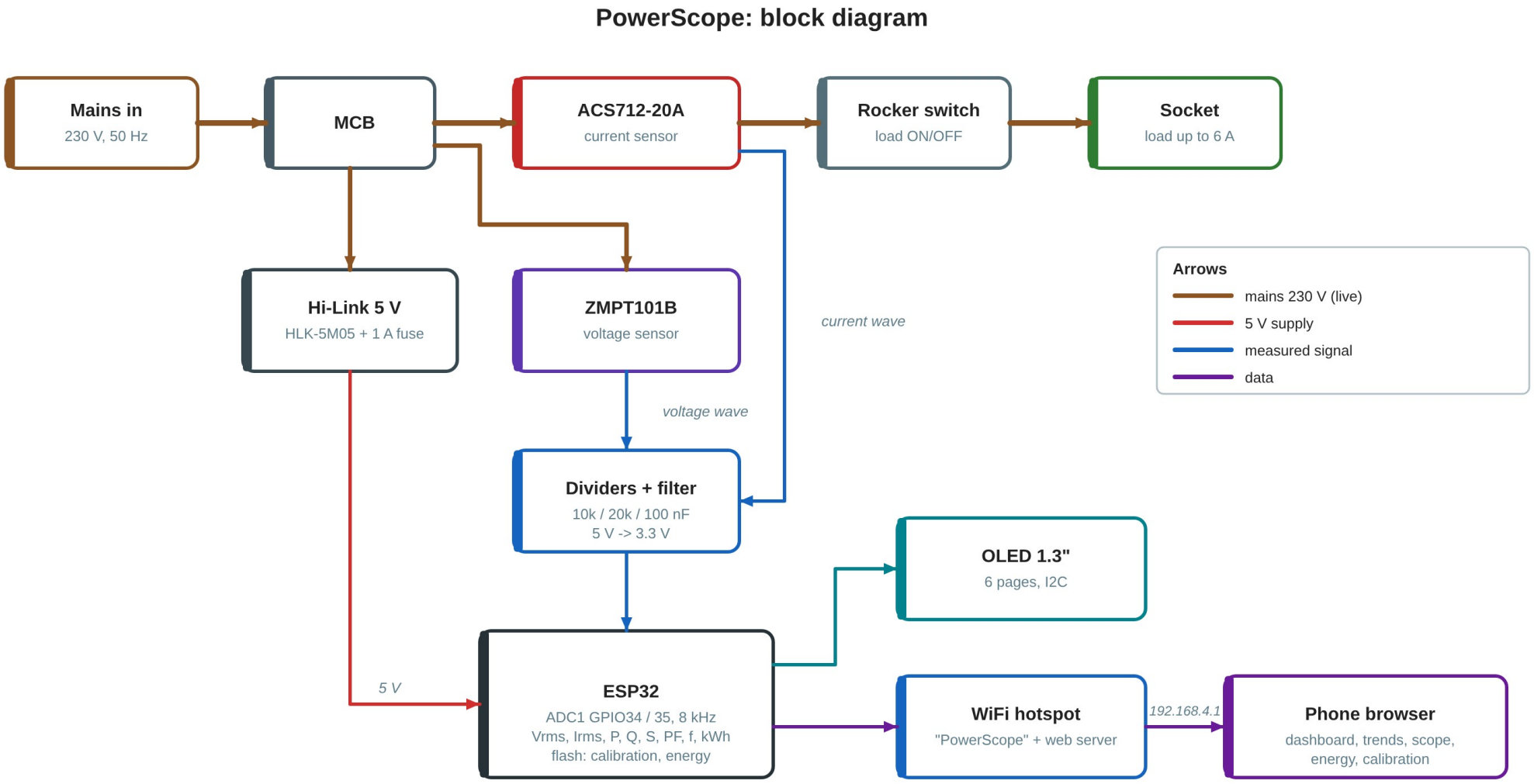


Figure 1. *PowerScope block diagram*

4. Components list

Table 2. Components

#	Component	Specification	Purpose
1	ESP32 DevKit	38-pin, ESP-WROOM-32, CP2102 USB	controller, WiFi, web server
2	Voltage sensor	ZMPT101B module, 5 V	measures mains voltage
3	Current sensor	ACS712-20A module, 5 V (100 mV/A)	measures load current
4	OLED display	1.3" SH1106, 128 x 64, I2C (4 pins)	local readout
5	Power supply	Hi-Link HLK-5M05, 230 V AC to 5 V 1 A	powers the electronics
6	Fuse + holder	1 A slow-blow, panel or inline	protects the Hi-Link input
7	Resistors	10 kΩ 1 % x2, 20 kΩ 1 % x2, 1/4 W	5 V to 3.3 V dividers
8	Capacitors	100 nF ceramic x2	divider noise filter
9	Capacitor	470 µF 16 V electrolytic	steadies the 5 V rail on WiFi bursts
10	MCB	single pole, modular	short-circuit and overload protection
11	Socket	6 A universal, modular	load socket
12	Rocker switch	modular, 6 A or more	manual load ON/OFF
13	Enclosure	modular plate + surface box, blank module	housing
14	Plug and cord	6 A 3-pin plug, 3-core flexible cord	mains input
15	Wire	1.5 sq mm for the load path; hook-up wire for low voltage	connections

5. Wiring

5.1 Pin map

Table 3. ESP32 connections

ESP32 pin	Goes to	Note
5V / VIN	5 V rail (Hi-Link +Vo)	no series diode; 470 µF across VIN and GND
GND	GND rail	one common ground for everything
3V3	OLED VCC	3.3 V output of the ESP32
GPIO34	voltage divider node (ZMPT101B OUT)	ADC1, input only
GPIO35	current divider node (ACS712 OUT)	ADC1, input only
GPIO21	OLED SDA	I2C data
GPIO22	OLED SCL	I2C clock

Table 4. Module connections

Module	Pin	Connect to
ZMPT101B	VCC / GND / OUT	5 V / GND / 10 kΩ to the GPIO34 node
ZMPT101B	L / N (input screw terminals)	live after the MCB and neutral
ACS712-20A	VCC / GND / OUT	5 V / GND / 10 kΩ to the GPIO35 node
ACS712-20A	IP+ / IP−	in series with the live wire: MCB out to IP+, IP− to the rocker switch
Divider nodes	–	20 kΩ to GND and 100 nF to GND at each node
OLED 1.3"	VCC / GND / SCL / SDA	3V3 / GND / GPIO22 / GPIO21
Hi-Link	AC L / AC N	live after the MCB through the 1 A fuse / neutral
Hi-Link	+Vo / −Vo	5 V rail / GND rail

5.2 Diagrams

Figure 2 shows every low-voltage connection on the real 38-pin pin order (used pins highlighted). Figure 3 shows the 230 V path.

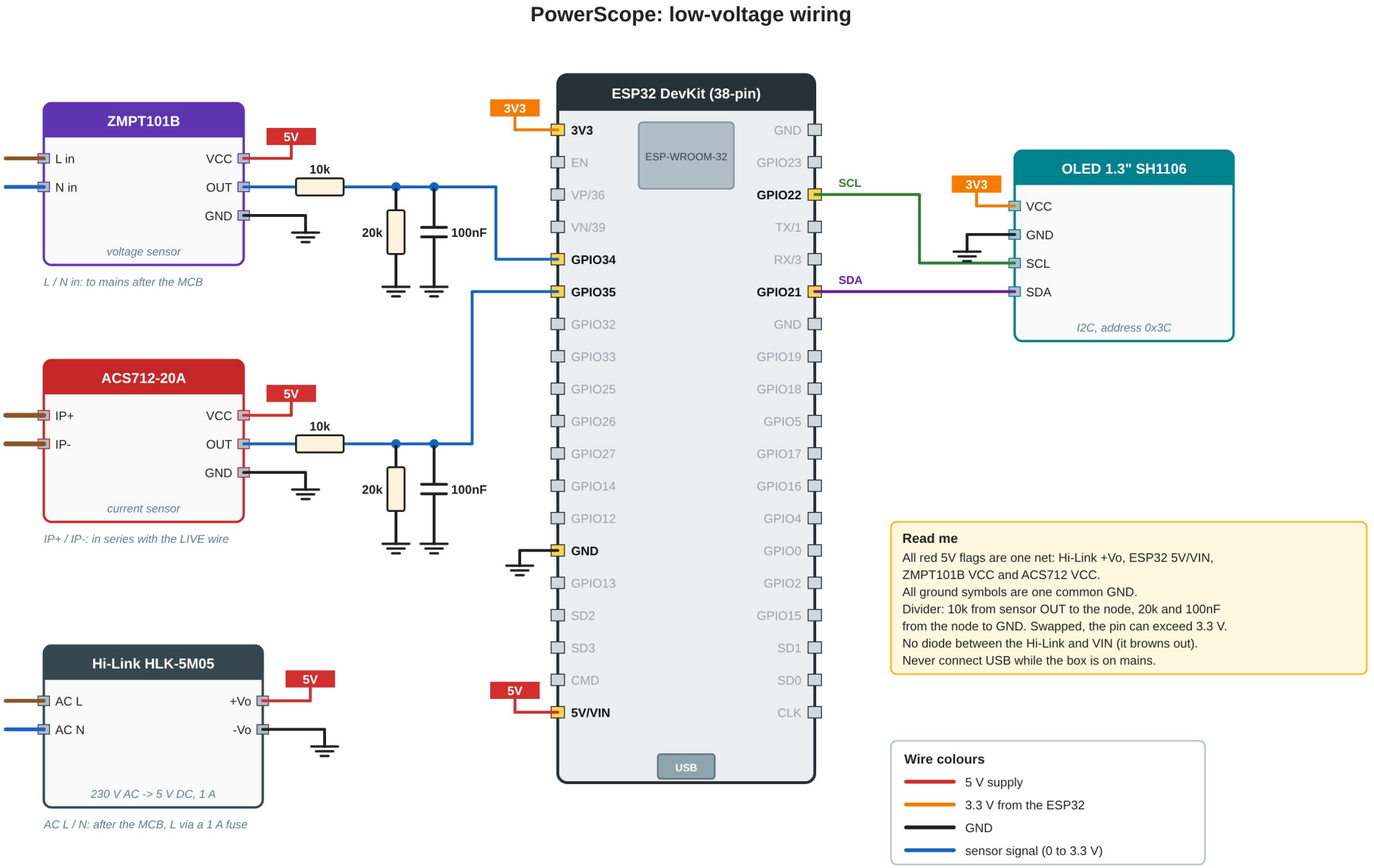
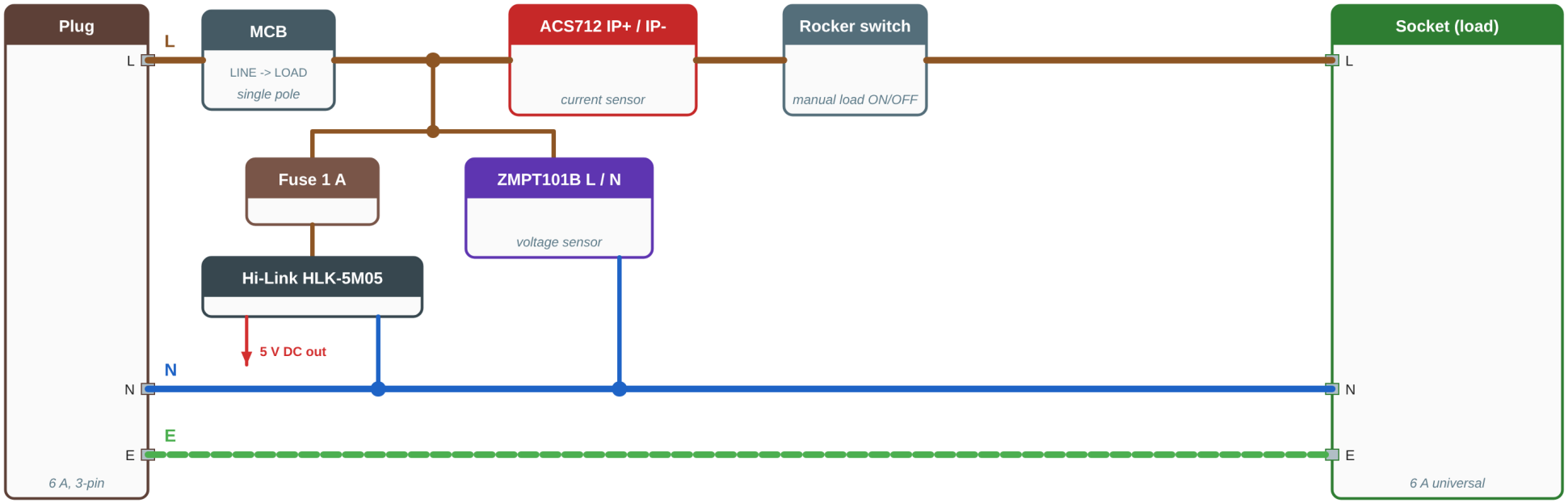


Figure 2. *PowerScope low-voltage wiring*

PowerScope: mains wiring (230 V)



Read me
Only the live wire goes through the MCB, the ACS712 and the rocker switch. Neutral and earth go straight from plug to socket.
Hi-Link and ZMPT101B are taken after the MCB. Test the wall socket for swapped live and neutral first. Use 1.5 sq mm wire for the load path, crimped lugs or ferrules in the terminals, and no bare metal outside the closed box.

Figure 3. *PowerScope mains wiring (230 V)*

6. Flashing and first setup

1. Install the Arduino IDE and the ESP32 board package (Espressif). Select board **ESP32 Dev Module**.
2. Set **Tools > Partition Scheme > Huge APP (3MB No OTA/1MB SPIFFS)**. Without it the sketch is too big.
3. Install libraries: **U8g2** (Library Manager), and **ESPAsyncWebServer** and **AsyncTCP** by ESP32Async (download each as a ZIP, then Sketch > Include Library > Add .ZIP Library).
4. Unzip the provided **PowerScope.zip**. It contains the folder **PowerScope** with **PowerScope.ino** and **web_index.h**; keep both files together in that folder. Open PowerScope.ino.
5. Connect the ESP32 by USB (**mains unplugged**), choose the COM port, and press Upload. Open the Serial Monitor at 115200 baud to see the readings.
6. On a phone, join WiFi **PowerScope** / **powerscope** and open **http://192.168.4.1**.

```
Board              ESP32 Dev Module
Partition Scheme   Huge APP (3MB No OTA/1MB SPIFFS)
Libraries          U8g2, ESPAsyncWebServer, AsyncTCP
Files              PowerScope.zip -> PowerScope/PowerScope.ino + PowerScope/web_index.h
Serial Monitor     115200 baud
```

7. Calibration and testing

Open the **Calibrate** section of the page and enter the PIN (**1234** until it is changed; five wrong tries lock it for 60 s). Every change is saved in flash and kept after a power cut.

1. **Voltage:** measure the socket with a multimeter (AC V). Type the reading under "Your multimeter reads, V" and press **Apply**. The shown voltage now matches the meter.
2. **Current:** plug in a kettle, iron or heater drawing at least 1 A (not a fan or charger). Measure the live wire with a clamp meter, type the reading under "Your clamp meter reads, A" and press **Apply**.
3. **Zero current:** unplug the load and press **Zero** (about 10 s). This measures the sensor's idle noise; afterwards an empty socket reads 0.00 A.
4. **Phase and direction (advanced):** with the kettle on, power must be positive and the power factor 1.00. If power is negative, press **Flip**. If the PF is below 0.99, step the phase shift with − / + (2.25° per step) until it reads 1.00.
5. Change the PIN under "PIN and defaults". **Restore defaults** puts back the factory calibration (the PIN is kept).

8. Display and web page guide

8.1 OLED pages

The OLED changes page every 5 s. The dots at the top right show which of the six pages is on.

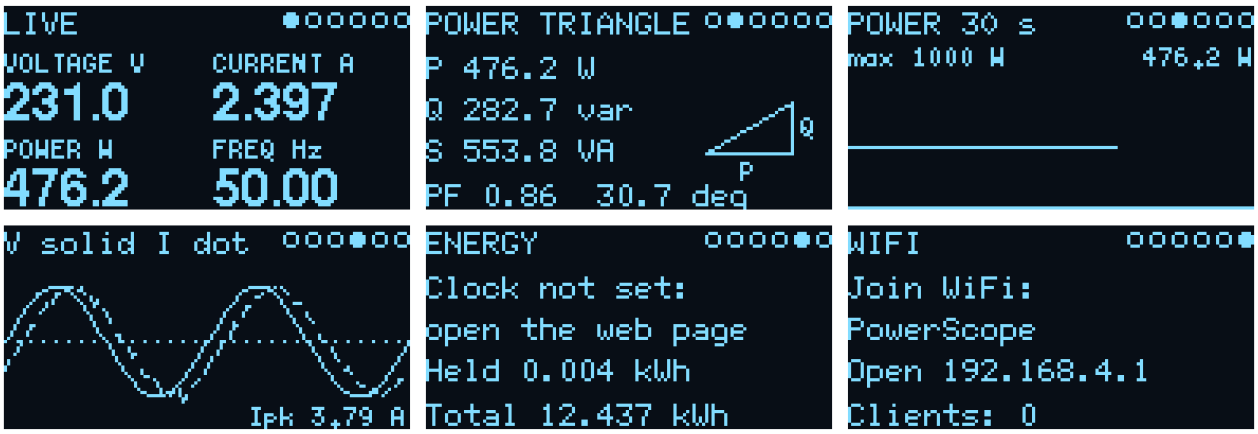


Figure 4. *The six OLED pages (rendered from the firmware with a 2.4 A load at PF 0.86)*

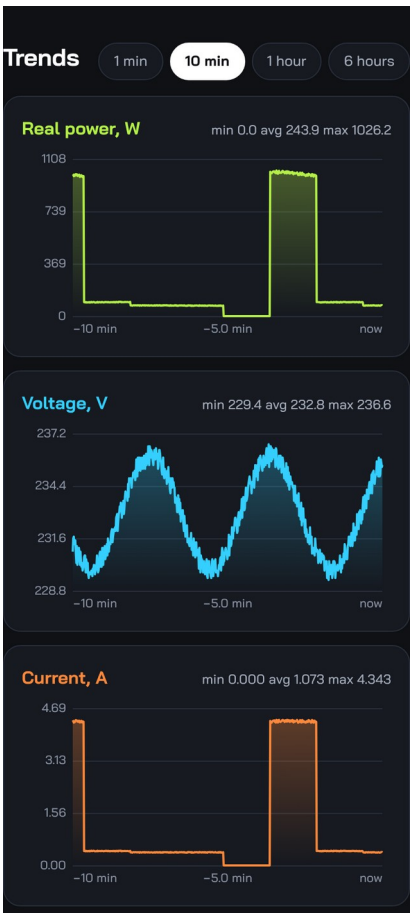
#	Page	Shows
1	LIVE	voltage, current, real power and frequency in large digits
2	POWER TRIANGLE	P, Q, S, power factor and phase angle, with the triangle drawn to scale
3	POWER 30 s	a 30 s graph of real power with the maximum
4	V solid I dot	two mains cycles of voltage (solid) and current (dotted), peak current
5	ENERGY	today and the total; "clock not set" until a phone opens the page
6	WIFI	WiFi name, address and the number of connected phones

8.2 Web dashboard

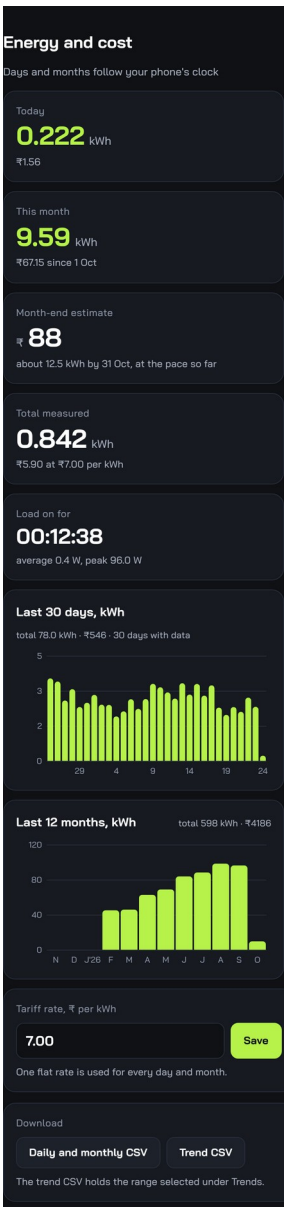
The page has five sections, reached from the bar at the top. If the page has never reached the device it shows demo data; once it has seen live data and loses contact, it shows "Offline" instead, so made-up numbers can never pass as real.



Live



Trends

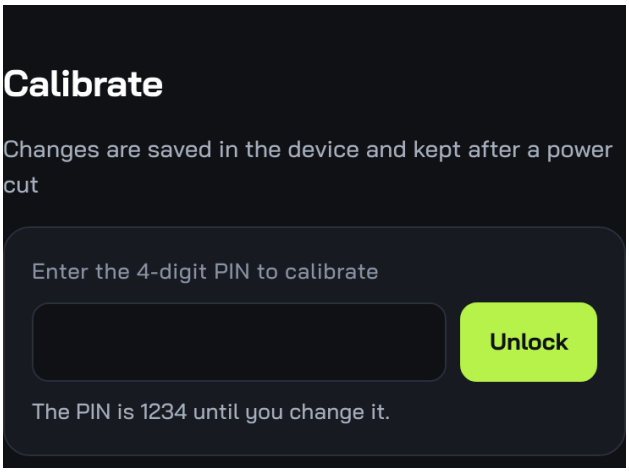


Energy and cost

Figure 5. *Dashboard sections (demo data)*



Scope



Calibrate (locked)

Section	What it does
Live	real power, S, Q, PF, phase angle, voltage, current, frequency, crest factor, power triangle; updated every second
Trends	power, voltage and current over 1 min, 10 min, 1 h or 6 h, with min / average / max
Scope	two mains cycles of voltage and current; pause to inspect; phase lag in ms and degrees
Energy and cost	today, this month, month-end estimate, total, load-on time, 30-day and 12-month charts, tariff, CSV download
Calibrate	PIN-protected calibration (Section 7)

9. Troubleshooting

Problem	Likely cause	Fix
Works on USB, OLED dark or resets on the Hi-Link	a diode between the Hi-Link and VIN drops the 5 V to about 3.6 V; the ESP32 browns out when WiFi starts	remove the diode, wire +Vo straight to VIN; add 470 µF; never connect USB on mains
Stuck on "starting WiFi"	supply too weak (see above)	measure VIN under load; it must stay above 4.5 V
OLED blank	SDA / SCL swapped, VCC on the wrong pin, wrong driver	SDA 21, SCL 22, VCC 3V3; for a 0.96" SSD1306 use the U8G2_SSD1306 line in the sketch
OLED picture shifted by 2 px or garbled	SSD1306 panel with the SH1106 driver (or the reverse)	swap the constructor line at the top of the sketch
Compile error "Ring was not declared" / "declared void"	an older copy of the sketch	use PowerScope.ino from the provided zip
"Sketch too big"	default partition scheme	Huge APP (3MB No OTA/1MB SPIFFS)
Power reads negative	current direction	Calibrate > Flip
Small loads (phone charger) read 0 A	below the noise floor (about 0.2 A with WiFi on)	expected; an ACS712-5A module reads lower
Current not 0 with nothing plugged in	zero not done for this unit	Calibrate > Zero with the load unplugged
Phone leaves the PowerScope WiFi	phone wants internet	PowerScope answers the internet checks; if a phone still drops, turn mobile data off
Days and months empty	no phone has opened the page since power-up	open the page once; held energy is added to that day
Readings jump when phones connect	WiFi bursts stretch a sampling window	those windows are dropped automatically; a few skipped updates are normal